Ran Li

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Summary

- Ph. D. in Imaging Science with 7+ years of experience in AI algorithm development, computer vision, Bayesian deep learning, and medical imaging research.
- Developed innovative strategies and implemented advanced algorithms for image analysis, achieving optimal segmentation results and outperforming classical deep learning-based methods.
- Led the design and implementation of algorithms for the analysis of medical images.
- Skilled in Python, C++, Matlab, Scikit, OpenCV, Pytorch, and ITK.
- Course works: Quantitative image analysis, Mathematics of Imaging Science, Computational Methods for Imaging Science, Biological Imaging Technology, Computer Vision

Experience

- **Research Scientist**
  EDDA Tech. *Feb. 2025-Now*
  - **Deep learning–based medical image processing for augmented-reality surgical navigation**: Develop and validate innovative deep learning-based methods for medical image processing to implement augmented-reality surgical navigation.
- **Research associate**
  Biomedical MR Center, Washington University in St. Louis *September, 2019 - January, 2025*
  - **Deep learning-based segmentation on carotid MR images**: 1. Implemented an innovative strategy that leveraged ex vivo high-resolution histopathology images to label in vivo MR images. 2. Attained optimal segmentation results by deploying a two-stage Bayesian U-Net, outperforming commonly used classical and deep learning-based methods for the same segmentation task.
  - **Deep learning-based inflammation biomarkers segmentation on PET-MR images** : 1. Developed a synthetic pipeline to address the absence of ground truth. 2. Designed a novel CNN backbone that outperformed both U-Net and various transformer-based segmentation methods for this specific task.
  - **Deep learning-based muscle metabolic quantitative analysis method**: 1. Created a medical physics-based pipeline for generating synthetic MR image datasets. 2. Developed a fully connected U-Net structure to produce precise and clear metabolic maps from MR images with artifacts, surpassing the performance of traditional quantitative calculation methods.
  - **Deep learning-based ROI selection method on ultrasound images**: 1. Utilized a Transformer U-Net to achieve automatic ROI selection. 2. Developed an automated image analysis pipeline, reducing manual analysis time for tendon ultrasound images by 99%.
- **Teaching Assistant of Mathematics in Imaging Science**
  Washington University in St. Louis *July 2021 - December 2021*
  - **Teaching**: Conducted 80 hours of individual and group tutoring sessions, utilizing presentations and code demonstrations to facilitate students' comprehensive understanding of learning concepts.
  - **Assignment Grading**: Graded assignment and exams covering subjects such as linear vector space theory, operator theory on Hilbert spaces, and applied functional analysis.
- **Research Assistant**
  Biomedical MR Center, Washington University in St. Louis *Sep 2017 - May 2019*
  - **Algorithm development for the customized software**: Led the design and implementation of algorithms within customized software packages for the analysis of MR images.
  - **Image denoising and registration toolkit development**: Compiled innovative and advanced solutions for denoising and registering medical images. Developed an executable toolkit using Python and PyQt5.
  - **Image 3D reconstruction**: Created an executable toolkit for 3D reconstruction of medical images using ITK, VTK, and PyQt5.

Education

- **Washington University in St. Louis** St. Louis, MO
  Ph.D Candidate - Imaging Science; GPA: 3.9 *September 2019 - May 2024*
- **Washington University in St. Louis** St. Louis, MO
  Master of Science - Electrical Engineering *September 2015 - May 2017*
- **Xi'an University of Technology** Xi'an, China
  Bachelor of Science - Electrical Engineering *September 2010 - July 2014*

Skills Summary

- **Programming Languages**: Python, Matlab, C++, C
- **Frameworks**: Pytorch, Scikit, OpenCV, Scipy, TensorFlow, QT,ANTS, ITK
- **Tools**: Git, Bash, Anaconda, Docker, Linux, Slurm
- **Technical**: Deep Learning, Machine Learning, Computer Vision, Image Processing, Algorithms, Medical Imaging

Journal Publications

- Nie X, Elvington A, Laforest R, Zheng J, Voller TF, Zayed MA, Abendschein DR, Bandara N, Xu J, **Li R**, Randolph GJ, Gropler RJ, Lapi SE, Woodard PK. 64Cu-ATSM Positron Emission Tomography/Magnetic Resonance Imaging of Hypoxia in Human Atherosclerosis. Circ Cardiovasc Imaging. 2020;13(1):e009791.
- Lu L, Eldeniz C, An H, **Li R**, Yang Y, Schindler TH, Peterson LR, Woodard PK, Zheng J. Quantification of myocardial oxygen extraction fraction: A proof-of-concept study. Magn Reson Med, 2021;85:3318-3325.
- Zheng J, **Li R**, Zayed MA, Yan Y, An H, Hastings MK, Pilot study of contrast free MRI reveals significantly impaired calf skeletal muscle perfusion in diabetes with incompressible peripheral arteries. Vasc Med. 2021;26:367-373.
- Zheng J, Sorensen C, **Li R**, An H, Hildebolt CF, Zayed MA, Mueller MJ, Hastings MK, Deteriorated regional calf microcirculation measured by contrast-free MRI in patients with diabetes mellitus and relation with physical activity. Diab Vasc Dis Res, 2021;18:14791641211029002.
- Wang K, Zhang W, Li S, Jin H, Jin Y, Wang L, **Li R**, Yang Y, Zheng J, Cheng J. Noncontrast T1 dispersion imaging is sensitive to diffuse fibrosis: A cardiovascular magnetic resonance study at 3T in hypertrophic cardiomyopathy. Magn Reson Imaging. 2022;91:1-8.
- Zheng J, **Li R**, Dickey EE, Yan Y, Zayed MA, Zellers JA, Hastings MK. Regional skeletal muscle perfusion distribution in diabetic feet may differentiate short-term healed foot ulcers from non-healed ulcers. Eur Radiol. 2023 Jan 31. doi: 10.1007/s00330-023-09405-6.
- **Li R**; Edalati M; Muccigrosso D; Lau JMC; Laforest R; Woodard PK, Zheng J. A Simplified Method to Correct Saturation of Arterial Input Function for Cardiac Magnetic Resonance First-Pass Perfusion Imaging: Validation with simultaneously acquired PET. Journal of Cardiovascular Magnetic Resonance, 2023;25(1):35.
- **Li R**, Zheng J, Saffitz JE, Woodard PK, Jha AK. Carotid atherosclerotic plaque segmentation in multi-weighted MRI using a two-stage neural network: Advantages of training with high-resolution imaging and histology. Frontiers in Cardiovasular Medicine, 2023.
- Wahidi R, Zhang Y, **Li R**, Zayed MA, Hastings MK, Zheng J. Quantitative Assessment of Peripheral Oxidative Metabolism with a new Dynamic 1H MRI: a Pilot Study in People with and without Diabetes Mellitus. J Mag Reson Imaging, 2023
- Zhang XL, Heo GS, Li A, Lahad D, Detering L, tao J, Gao XF, Zhang XH, Luehmann H, Sultan D, Lou L, Venkatesan R, **Li R**, Zheng J, Amrute J, Lin CY, Kopecky BJ, Gropler RJ, Bredemeyer A, Lavine K, Liu YJ, Development of a CD163 Targeted PET Radiotracer Imaging Resident Macrophages in Atherosclerosis. J Nucl Med. 2024, accepted.
- Zellers J, **Li R** Vaidya R, Lohse K, North A, Cui S, Houston B, Chen M, Zheng J, Baxter J, PhD. Effect of scanning parameters on ultrasound shear wave elastography variability in tendon. Knee Surgery, Sport Traumatology, Arthroscopy ,2024
- **Li R**, Eldeniz C, Wang K, Nguyen N, Schindler TH, Huang Q, Peterson LR, Yang Y, Yan Y, Cheng J, Woodard PK. Quantification of myocardial oxygen extraction fraction on noncontrast MRI enabled by deep learning. Radiology Advances. 2024 Nov;1(4):umae026.
- Huang Q, Tang H, Wang K, **Li R**, Eldeniz C, Nguyen N, Schindler TH, Peterson LR, Yang Y, Yan Y, Cheng J. Model-based self-supervised learning for quantitative assessment of myocardial oxygen extraction fraction and myocardial blood volume. Magnetic resonance in medicine. 2025 May 5.

Conference Abstract

- Edalati MA, Hastings MK, Mohamed ZA, Muccigrosso D, **Li R**, Mueller MJ, Zheng J, MRI Methods for Exercise-based Perfusion Assessment of Diabetic Feet with Ulcers. Annual Conference of International Society of Magnetic Resonance in Medicine, June, 2018, Paris, France
- Zheng J, Gharabaghi S, **Li R**, Chen YS, An H, Haacke EM, Zayed MA, Hastings MK. Pilot contrast free MRI reveals significantly impaired calf skeletal muscle perfusion in diabetes with incompressible peripheral arteries. Annual Conference of International Society of Magnetic Resonance in Medicine, May, 2021, Paris, France (virtual).
- Wahidi R, Zhang Y, **Li R**, Zayed MA, Hastings MK, Zheng J. Quantitative assessment of peripheral metabolism and perfusion with dynamic 1H MRI inhealthy and diabetic individuals. Annual Conference of International Society of Magnetic Resonance in Medicine, May, 2022, London, UK.

- **Li R** Zheng J, Woodard PK, Jha AK. Carotid atherosclerotic plaque segmentation in multi-weighted MRI using a two-stage neuralnetwork model. Annual Conference of International Society of Magnetic Resonance in Medicine, May, 2022, London, UK.
- Wang K, Zhang WB, Li S, Jin HR, Jin Y, Wang Li, **Li R**, Yang Y, Cheng JL, Zheng J, An J. T1 dispersion imaging on 3T MRI detects diffuse fibrosis in subtypes of hypertrophic cardiomyopathy. Annual Conference of International Society of Magnetic Resonance in Medicine, May, 2022, London, UK.
- Mei BC, **Li R**, Eldeniz C, Schindler T, Peterson LR, Woodard PK, Zheng J. Deep Learning Assisted CMR Myocardial Oxygen Extraction Fraction Measurements. Annual Conference of Society for Cardiovascular Magnetic Resonance, Jan, 2023, San Diego, CA.
- Wang J. **Li R**, Fu X, Yang Y, Sun XJ, Shu S, Zhao J, Kong XC, Zheng J. Myocardial oxygen extraction fraction moderately correlates with fibrosis burden in patients after heart transplant: a MR metabolism study. Annual Conference of International Society of Magnetic Resonance in Medicine, June, 2023, Toronto, Canada.
- **Li R**, Eldeniz C, Schindler TH, Peterson LR, Woodard PK, Jie Zheng. Deep learning assisted quantification of myocardial oxygen extraction fraction. Annual Conference of International Society of Magnetic Resonance in Medicine, June, 2023, Toronto, CA. (Travel Awarded issued by the Conference).
- **Li R**, Kaminski O, Eldeniz C, Schindler T, Peterson L, Woodard P, Zheng J. Artificial intelligence-based quantification of myocardial oxygen extraction fraction and blood volume in health with CMR: reproducibility and distribution. Annual conference of Society for Cardiovascular Magnetic Resonance, Jan, 2024 London, UK (In person presentation)
- **Li R**, Zheng J.Quantification of myocardial blood flow in vivo with deep learning-assisted cardiac arterial spin labeling (DeepCASL) MRI: towards validation by microsphere approach Annual Conference of International Society of Magnetic Resonance in Medicine, May, 2024, Singapore
- **Li R**,Eldeniz C, Schindler T, Peterson L, Woodard P, Zheng J. Artificial intelligence-based quantification of myocardial oxygen extraction fraction and blood volume in health: reproducibility, sex, and homogeneity

  Annual Conference of International Society of Magnetic Resonance in Medicine, May, 2024, Singapore

Patent

- “Carotid Plaque Segmentation Using a Trained Neural Network,” US Provisional Application 63/399,325, filed August 19, 2022 (**Li R.**, Jha A., Zheng J., Woodard P.)